# Chapter 2: Linearization and Log-Linearization

Gauti B. Eggertsson, Brown University, Updated June 2, 2020

## 1 Overview

As a Ph.D. student at Princeton in the late 90s I attended a class that I believe was in the engineering department. The title of the class was non-linear dynamics, or something of that sort. I figured that surely those tools would be helpful to study economic phenomena, which appear to be highly non-linear, such as economic crises.

The professor started the class by saying that he recalled being student and thinking that it made no sense to approximate non-linear dynamics with a linear or log-linear approximation. He then added, "and now, all these years later, I continue to be surprised that in practice in my field, aerodynamics, they work extremely well." I dropped the class soon afterward, figuring that if linear approximations worked well in aerodynamics, they would probably be sufficient for economics as well.

I have myself had the same experience as that professor. In practice, I have found linear and log-linear approximation works surprisingly well for the applications I have worked on, even when considering extreme events such as the Great Depression or hyperinflations, both of which will feature later on. For this reason, I will focus these notes on using linear and log-linear approximations. This chapter outlines the theory behind them.

A macroeconomics model typically involves a set of non-linear equations that are being "shocked" by some exogenous shocks (think of coin tosses). The object of interest is typically to understand how the system responds to the exogenous shocks, e.g. monetary expansion/contraction or fiscal expansion/contraction.

A very common way of characterizing the solution is to:

1. Find a steady state (constant solution) for the system of equations, which is the solution in the absence of any shock.

2. Linearize the system around the steady state via first-order Taylor expansion so that the result is a set of linear equations in the endogenous variables and the shocks.

3. Solve the set of linearized equations so that one can write all the endogenous variables as function of the exogenous shocks.

This approach simplifies the calculation considerably, making the models easy to simulate, estimate and, perhaps no less importantly, interpret. The key behind the tractability is that having a set of stochastic linear equations means you can apply the powerful toolbox of linear algebra to work with the model.

## 2 What is a Linearization?

Consider the function $f(x)$, which we assume to be differentiable. A first-order Taylor approximation of this function around the point $x^*$ is:

$$f(x) \approx f(x^*) + f'(x^*)(x - x^*),$$

where $f'(x^*)$ is the derivative of the function $f(x)$ evaluated at the point $x^*$. We will use Taylor expansion to approximate models, that is, we will replace non-linear equations with linear approximations. It is easiest to see how this works via examples.

Consider the function $y = g(x) = x^2$. What is the linear approximation of this function around the point $x^* = 1$? Let us call the approximated function $\hat{g}(x)$. Then:

$$g(x) \approx \hat{g}(x) = g(x^*) + g'(x^*)(x - x^*) = 1^2 + 2(x - 1) = 2x - 1$$

and, as you can see, the function is now linear in $x$. How good is this approximation? Consider the point $x = 1.1$:

$$\hat{g}(x) = 2x - 1 = 1.2$$

Meanwhile, the true value of $g(x)$ at 1.1 is:

$$g(x) = x^2 = (1.1)^2 = 1.21$$

so the approximation does well. The example is shown in Figure (1). The approximated function $\hat{g}(x)$ is the red line while the true function is the blue parabola. The red line is tangent to the blue line at the point we expand around, i.e. 1. The figure shows an important point: while the approximation works pretty well locally – meaning close to 1 –, it will get progressively worse as we move away from 1. Consider, for example, the case when $x = -1$. Then the approximate function says $\hat{g}(-1) = -3$, while the true function says $g(1) = 1$ –

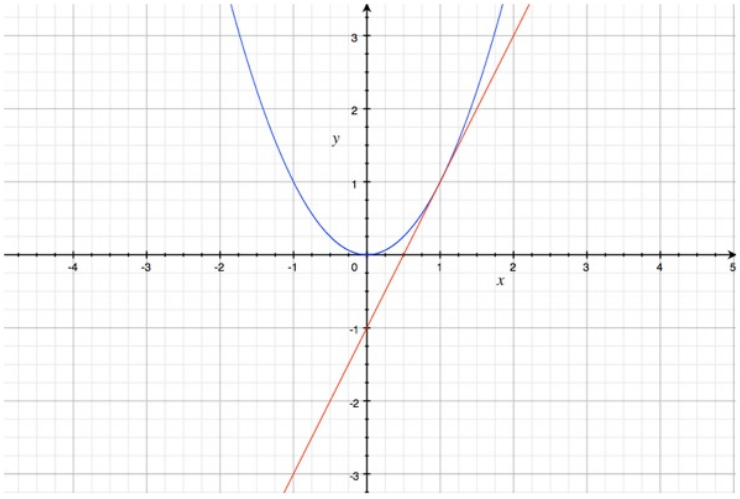



Figure 1: Linear Approximation of a Parabola

a poor approximation indeed! This highlights an important principle: the approximation of macro models via first-order approximations is only correct local to the steady state, and thus it becomes important to check how plausible that assumption is in practice (answer: to a surprising degree, linear approximation work quite well in many applications).

Why is it useful to use $\hat{g}(x)$ rather than $g(x)$? Because the function $\hat{g}(x)$ is linear in $x$, while $g(x)$ is not. This makes it easier to solve models, especially those that involve several variables and a large number of equations, as we can use many powerful tools of linear algebra. A Taylor expansion in several variables is a straightforward extension. Consider, for example, $f(x, y)$. The first-order Taylor expansion is given by:

$$f(x, y) \approx f(x^*, y^*) + f_x(x^*, y^*)(x - x^*) + f_y(x^*, y^*)(y - y^*),$$

where $f_x(x^*, y^*)$ denotes the partial derivative of $f(x, y)$ with respect to $x$ evaluated at $(x^*, y^*)$.

As an example, consider the function:

$$g(x, y) = x^2 + \frac{10}{y}$$

Let us expand this function around the point $(x^*, y^*) = (1, 10)$. This gives:

$$\begin{aligned} g(x,y) &\approx \hat{g}(x^*,y^*) = g(x^*,y^*) + g_x(x^*,y^*)(x-x^*) + g_y(x^*,y^*)(y-y^*) \\ &= 2 + 2(x-1) - \frac{1}{10}(y-10) = 1 + 2x - \frac{y}{10} \end{aligned}$$

Consider how well this approximation does for $x, y = (1.1, 10.1)$. Then, $\hat{g}(1.1, 10.1) = 1 + 2 * 1.1 - \frac{10.1}{10} = 2.19$, while $g(1.1, 10.1) = 1.1^2 + \frac{10}{10.1} = 2.2$. You can confirm that the approximation becomes worse the further we move away from the expansion point.

# 3 Linearizing a Model

The rationale for using linearization in practice is that we are studying models that return, in the absence of shocks, to a well defined "steady state" – where the steady state is the solution of the model in absence of any shocks. The linear approximation is typically done around this steady state. If the shocks are small, the linear approximation will be good. As we will see below, the convention is to write the variables of the model in terms of deviation from this steady state.

Consider the following model which we will consider in the coming lectures, where $G_t$ is government consumption, $Y_t$ output, $r_t$ the real interest rate, $C_t$ consumption, $b_t$ debt and $\beta$ and $\sigma$ are parameters. We make the assumption that $G_t = \bar{G}$ in steady state (even if it may vary stochastically) and, for simplicity, that taxes are *always* set so that debt is constant at $b_t = \bar{b}$. These two assumptions imply that the equilibrium is defined by the following four conditions (where the first comes from the government budget constraint):

$$T_t = r_{t-1}\bar{b} + G_t \tag{1}$$

$$Y_t = C_t + G_t \tag{2}$$

$$C_t^{-\sigma} = \beta(1 + r_t)E_t C_{t+1}^{-\sigma} \tag{3}$$

$$\lim_{T\to\infty} E_t \beta^T C_T^{-\sigma} \xi_T b_T = 0 \tag{4}$$

where the source of uncertainty is given by $Y_t$. We can now characterize an approximate solution by following steps 1-3. The first step is to solve for the steady state – the solution of the model in the absence of shocks, i.e. when

$Y_t = \bar{Y}$. In this case we have $C_t = \bar{C} = \bar{Y} - \bar{G}$ by the second equation, the third equation implies that $1 + r_t = 1 + \bar{r} = \beta^{-1}$, so that $\bar{r} = \beta^{-1} - 1$, while $T_t = \bar{T} = \bar{r}\bar{b} + \bar{G}$.

Let us use the notation that $dx_t \equiv x_t - \bar{x}$, i.e., $dx$ denotes the deviation of a variable from its steady state. The Taylor approximation says that

$$f(x_t, y_t) \approx f(\bar{x}, \bar{y}) + f_x(\bar{x}, \bar{y})dx_t + f_y(\bar{x}, \bar{y})dy_t \tag{5}$$

Since the first two equations are already linear, we can rewrite them as (subtracting each equation from itself evaluated in steady state):

$$dT_t = \bar{b}dr_{t-1} + dG_t$$

$$dY_t = dC_t + dG_t$$

The third equation, however, needs to be linearized explicitly using formula (5), starting with the left side and then following up with the right-hand side:

$$C_t^{-\sigma} \approx \bar{C}^{-\sigma} - \sigma\bar{C}^{-\sigma-1}dC_t$$

$$\beta(1 + r_t)E_tC_{t+1}^{-\sigma} \approx \beta(1 + \bar{r})\bar{C}^{-\sigma} + \beta\bar{C}^{-\sigma}dr_t - \sigma\beta(1 + \bar{r})\bar{C}^{-\sigma-1}E_tdC_{t+1}$$

Equating the two sides, noting that the constant terms will cancel out, that $1 + \bar{r} = \beta^{-1}$ and dividing both sides by $-C^{-\sigma}$ we obtain:

$$\sigma\bar{C}^{-1}dC_t = \sigma\bar{C}^{-1}E_tdC_{t+1} - \beta dr_t,$$

which completes the linearization.

Typically, macreconomists express the linearized equations in terms of the percentage deviation of each of the variables from its steady state, denoted by $\hat{x}_t \equiv \frac{dx_t}{\bar{x}} = \frac{x_t - \bar{x}}{\bar{x}}$ (note that we define $r_t = \frac{dr_t}{1+\bar{r}}$). Using this notation, we obtain:

$$\hat{T}_t = \frac{\bar{b}}{\bar{T}}\beta^{-1}\hat{r}_{t-1} + \frac{\bar{G}}{\bar{T}}\hat{G}_t \tag{6}$$

$$\hat{Y}_t = \frac{\bar{C}}{\bar{Y}}\hat{C}_t + \frac{\bar{G}}{\bar{Y}}\hat{G}_t \tag{7}$$

$$\hat{C}_t = E_t\hat{C}_{t+1} - \sigma^{-1}\hat{r}_t \tag{8}$$

An approximate equilibrium is thus a collection of stochastic processes for $\{\hat{C}_t, \hat{T}_t, \hat{r}_t\}$, for a given exogenous path for $\{\hat{Y}_t, \hat{G}_t\}$, which solves the three equations above (we will not write the transversality condition explicitly, as it will always be satisfied as long as the solution to the model is bounded). A solution to this model expresses the endogenous variables in terms of the

exogenous ones, i.e. $\hat{r}_t, \hat{T}_t, \hat{C}_t$ in terms of $\hat{G}_t$ and $\hat{Y}_t$ in the example above. To see why it is useful to approximate the equilibrium in this way, consider the following process for $\hat{Y}_t$ and $\hat{G}_t$:

$$\hat{Y}_t = \rho \hat{Y}_{t-1} + \epsilon_t$$

$$\hat{G}_t = \gamma \hat{G}_{t-1} + u_t,$$

which now allows us to solve the model in closed form. The value of $\hat{C}_t$ follows directly from (7):

$$\hat{C}_t = \frac{\bar{Y}}{\bar{C}} \hat{Y}_t - \frac{\bar{G}}{\bar{C}} \hat{G}_t \tag{9}$$

To find the solution for $\hat{r}_t$, note that using (7) we find that $E_t \hat{C}_{t+1} = \frac{\bar{Y}}{\bar{C}} E_t \hat{Y}_{t+1} - \frac{\bar{G}}{\bar{C}} E_t \hat{G}_{t+1} = \rho \frac{\bar{Y}}{\bar{C}} \hat{Y}_t - \gamma \frac{\bar{G}}{\bar{C}} \hat{G}_t$. Using (8), then:

$$\hat{r}_t = -\sigma(1-\rho) \frac{\bar{Y}}{\bar{C}} \hat{Y}_t + \sigma(1-\gamma) \frac{\bar{G}}{\bar{C}} \hat{G}_t \tag{10}$$

The expression for $\hat{T}_t$ can be found by substituting this into (6) and obtain:

$$\hat{T}_t = -\frac{\bar{b}}{\bar{T}} \beta^{-1} \sigma \left[ (1-\rho) \frac{\bar{Y}}{\bar{C}} \hat{Y}_{t-1} - (1-\gamma) \frac{\bar{G}}{\bar{C}} \hat{G}_{t-1} \right] + \frac{\bar{G}}{\bar{T}} \hat{G}_t \tag{11}$$

Equations (9)-(11) constitute an approximated solution to the model, i.e. we express the endogenous variables $\hat{C}_t$, $\hat{r}_t$, $\hat{T}_t$ as a function of the exogenous shocks $\hat{Y}_t$ and $\hat{G}_t$, so it is straightforward to compute impulse responses, forecast the endogenous variables and so on. Note that this solution is in closed form, while the solution to the non-linear equations is typically numerical unless one makes special assumptions about functional forms and special stochastic processes driving the exogenous variables. The key advantage of the current approach is that it is relatively computationally "cheap", so that it can be applied to large systems of equations, which can then be solved in the same way we did above, with several computer packages available that do most of the work for you (including the linearization).

# 4 Log-linearization

The macroeconomic literature will generally not do a linearization of the sort I did above but instead will rely on *log-linearization*, where the hatted variables are now defined as $\hat{x}_t = \log x_t - \log \bar{x}$. Observe that approximation via percentage deviation from steady state and log-linearization are equivalent to the

first order. To see this, do a first order Taylor approximation of the function $f(x) = log(x) - log(x^*)$ around the point $x^*$. It is:

$$f(x) = log(x) - log(x^*) \approx f(x^*) + f'(x^*)(x - x^*) =$$

$$= \log(x^*) - \log(x^*) + \frac{1}{x^*}(x - x^*) = \frac{x - x^*}{x^*}$$

Below, I show how log-linearizing is done in practice in many textbooks, just to help you confirm that the same results apply. I will take the consumption Euler Equation (3) as an illustrative example. If we define $\hat{C}_t \equiv \log C_t - \log \bar{C}$ and $\hat{r}_t \equiv \log(1 + r_t) - \log(1 + \bar{r})$, then it follows that $C_t = \bar{C}e^{\hat{C}_t}$ and $1 + r_t = (1 + \bar{r})e^{\hat{r}_t}$. Substitute this into (3) to obtain:

$$(\bar{C}e^{\hat{C}_t})^{-\sigma} = \beta(1 + \bar{r})e^{\hat{r}_t}E_t(\bar{C}e^{\hat{C}_{t+1}})^{-\sigma} \tag{12}$$

We can now do a first order Taylor expansion in $\hat{C}_t$, $\hat{r}_t$ and $\hat{C}_{t+1}$ around 0 (since the hatted variables are in deviation from steady state, expanding around 0 means we are expanding around the steady state), to obtain the following for the left-hand side and then the right-hand side of the equation:

$$(\bar{C}e^{\hat{C}_t})^{-\sigma} \approx (\bar{C}e^0)^{-\sigma} - \sigma(\bar{C}e^0)^{-\sigma-1}\bar{C}e^0(\hat{C}_t - 0) = \bar{C}^{-\sigma} - \sigma(\bar{C})^{-\sigma}\hat{C}_t$$

$$\begin{aligned} & \beta(1 + \bar{r})e^{\hat{r}_t}E_t(\bar{C}e^{\hat{C}_{t+1}})^{-\sigma} \\ \approx\ & \beta(1 + \bar{r})e^0(\bar{C}e^0)^{-\sigma} + \beta(1 + \bar{r})e^0(\bar{C}e^0)^{-\sigma}\hat{r}_t - \sigma e^0(\bar{C}e^0)^{-\sigma-1}\bar{C}e^0E_t\hat{C}_{t+1} \\ =\ & \beta(1 + \bar{r})(\bar{C})^{-\sigma} + \beta(1 + \bar{r})(\bar{C})^{-\sigma}\hat{r}_t - \sigma(\bar{C})^{-\sigma}E_t\hat{C}_{t+1} \end{aligned}$$

Equating these two, rearranging and dividing by $(\bar{C})^{-\sigma}$ yields (8).

In principle, it does not matter which way you do the linearization, i.e. rewriting in terms of logs to obtain (12) and do a first order Taylor expansion, or directly do a first order Taylor expansion as we did in the last section. Since the two approaches yield the same coefficients in the linearized equations, you can do it either way, and then define $\hat{x}_t$ as either log deviation or percentage deviation. My own preference is to use the former approach. This is mainly because, in some cases, model variables have a steady state of 0 (for example zero government debt in some of the examples we will see), in which case the log of the variable in steady state is not defined (you can't take a log of 0!). But this is just a matter of taste, not logic.

# 5 Percentage vs log linearization: Which one is more accurate?

What gives more accurate approximation to the exact model equations, defining the variables in terms of log deviation or percentage deviation from steady state? Even if the two are equivalent to first order, which approach you use will result in different approximation errors. How big they are depends on the nature of the model. The reason why macroeconomists tend to use log-linearizations is that, in practice, log-linearization tends to approximate commonly used models better, because many equations in macroeconomics are in exponential form, where log-linearization may provide an exact solution. Examples help clarify this point. First, I present an example in which defining the variable in terms of percentage deviation works better, as the model is already linear; then, an example in which a log deviation works better, as the model is already exponential. The examples are, by construction, the most simple ones I could think of, to highlight the differences between the two approaches.

**Example where percentage deviation works better**

Consider the following model, in which $x$ and $y$ are the exogenous variables, and $z$ the endogenous one. The model to be "approximated" is:

$$z = x + y.$$

Denote the point around which the model is approximated by $(x^*, y^*)$. Approximating this equation yields:

$$\hat{z} = \frac{x^*}{z^*}\hat{x} + \frac{y^*}{z^*}\hat{y}, \tag{13}$$

where $\hat{z} \equiv \frac{z-z^*}{z^*}$, for an approximation using percentage deviation and $\hat{z} \equiv \log z - \log z^*$, for a log-linear approximation. Note that there is no approximation error if we rely on percentage deviation from steady state. There is an approximation error, however, in the case of the log-linear approximation. To see this, consider, as an example, the expansion point of $(x^*, y^*) = (1, 1)$ and let us evaluate this expression in a new equilibrium in which $x = 4$ and $y = 1$. In this case, obviously, $z = 5$ in the exact model. Since the approximation relying on percentage deviation from steady state is exact, the answer is again 5 without any approximation error, i.e. $\hat{z} = \frac{1}{2} \cdot 3$ according to (13), so that $z = z^* + z^* \cdot \hat{z} = 2 + 2 \cdot \frac{1}{2} \cdot 3 = 5$. The log-linear approximation, however, gives rise to an approximation error. To see this, the approximated equation is $\log z - \log z^* = \frac{1}{2} \cdot \log 4$ according to (13), so that $\log z = \log 2 + \log 4^{1/2} = \log 2^2$, i.e. $z = 4$. Hence, log-linearizing a linear constraint will in general lead to an

approximation error that becomes larger as we move further away from the point around which we expand.

**Example where log-linearization works better**

Consider the example where $z = g(x) = x^2$. Here, a log-linearization is exact. To see this, take logs of both sides to get $\log z = 2 \log x$. Now subtract from both sides the expression evaluated at the point we want to "expand" around to obtain $\hat{z} = 2\hat{x}$, where now $\hat{x} = \log x - \log x^*$. Note that now, starting from the steady state of $x^* = 1$, if we consider $x = -1$, with the log-linearization we will get the correct answer 1, in contrast with the linear approximation where the result is $-3$ and the approximation error is severe. This is a useful example because many economic models are exponential (e.g. the Cobb-Douglas production function is exponential) and thus the log-linearization will yield an exact solution to the equations of the model that are in exponential form. This explains why in practice log-linearization tends to work better, even if the example before suggests that whether this is true needs to be evaluated on case by case basis. Generally, when you have a large number of constraints in additive form, then percentage deviation is better, while if you have large number constraint with multiplicative constraints, then log-linearization works better.

# 6 Higher order approximations

There is in principle no reason to only approximate models to the first order. Taylor approximation can be of arbitrarily high order and one can often obtain a more accurate approximation by doing a higher order approximation (even if there is no guarantee that the higher order approximation will converge to the truth, this depends on conditions that need to be checked on case by case basis). A second order approximation yields:

$$f(x) \approx f(x^*) + f'(x^*)(x - x^*) + \frac{f''(x^*)}{2}(x - x^*)^2$$

As an example, a second order approximation of $y = g(x) = x^2$ around the point $x = 1$ yields:

$$y \approx 1 + 2(x - 1) + (x - 1)^2 = x^2,$$

i.e. one obtains the exact solution. There are methods that allow you to solve models using higher order approximation, which can be important when one wants to think about time varying uncertainty or "risk premia". These methods

critically depend upon the model being "well behaved", in particular being differentiable.

# 7 Mathematical theory (to be added)

The mathematical theory behind linear approximation is essentially an application of the implicit function theorem and the inverse function theorem, see for example a relatively intuitive discussion in Woodford (2003), Interest and Prices, Addendum to Chapter 2, A3.

# 8 Problems

**Problem 1**

There is no economic in this question, I am only asking you to do a simple linearization and compute an impulse response. Consider the following simple economy, where $Y_t$ is output, $K_t$ capital, $I_t$ investment, $L_t$ is labor:

$$Y_t = K_t^{\alpha} L_t^{1-\alpha}$$

$$K_t = \delta K_{t-1} + I_t$$

$$I_t = \bar{I} + \epsilon_t^I$$

$$L_t = \bar{L} + \epsilon_t^L$$

where $0 < \delta < 1$ and $0 < \alpha < 1$. The shocks $\epsilon_t^i$, where $i = L, I$, are all iid with mean 0 and $\bar{I}$ and $\bar{L}$ are steady state values for each shock.

i. Do a linear approximation of this system around the constant solution $Y = \bar{Y}$, $K_t = \bar{K}$, $I_t = \bar{I}$, $L_t = \bar{L}$. Define each variable in terms of percentage deviation from steady state, e.g. $\hat{K}_t = \frac{K_t - \bar{K}}{\bar{K}}$.

ii. Compute the impulse response of output with respect to $\epsilon_t^I, \epsilon_t^L$, i.e., compute the derivatives $E_t \frac{\partial Y_{t+j}}{\partial \epsilon_t^i}$ for all $j \geq 0$ and $i = L, I$.

iii. Do a log-linear approximation defining each variables as e.g. $\hat{K}_t = \log K_t - \log \bar{K}$. How do the results compare and which approximation do you think works better for approximating each equation? You can use a parametrization to evaluate this.